Universität Basel

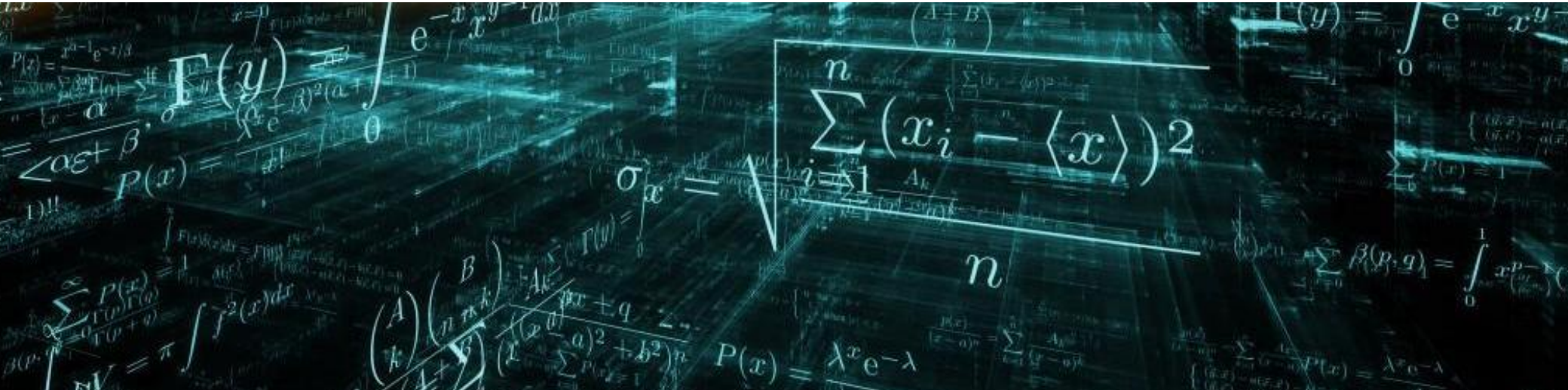
Computer Science / 15731-01 / 2026

Multimedia Retrieval

Chapter 9: Content Analysis & Metadata

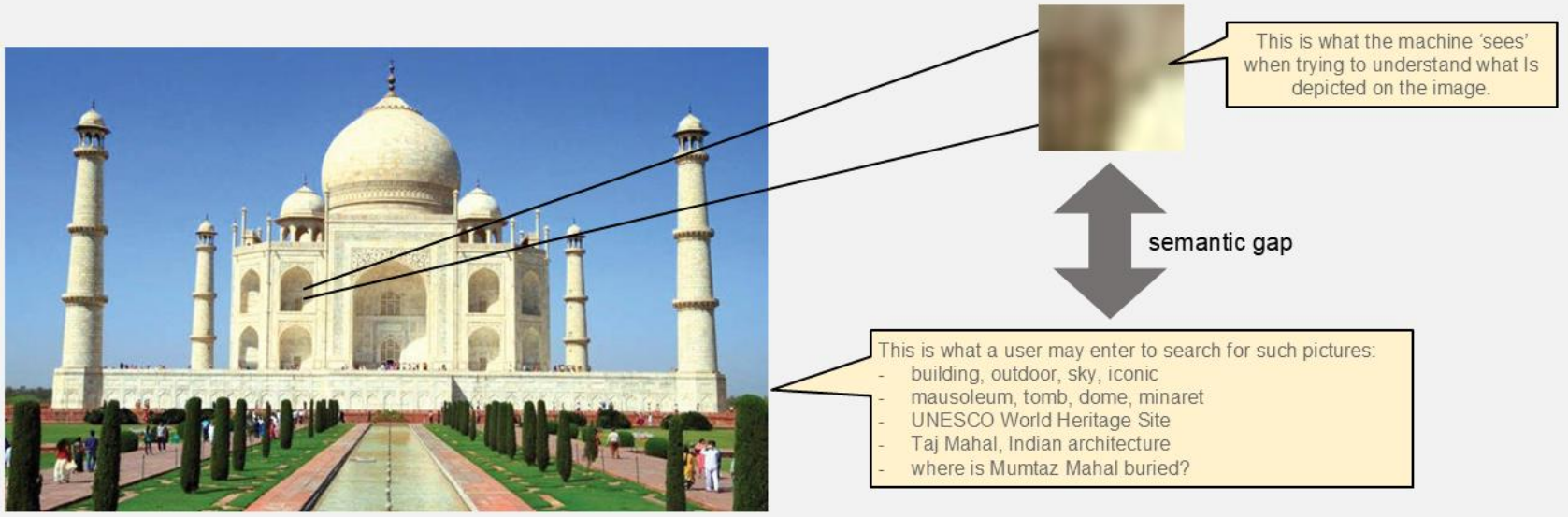
Dr. Roger Weber, roger.weber@gmail.com

9.1 The Semantic Gap
9.2 Metadata and Its Limits
9.3 Automated Feature Extraction
9.4 Multimodal Transformers
9.5 Time-Based Media and Specialized Solutions



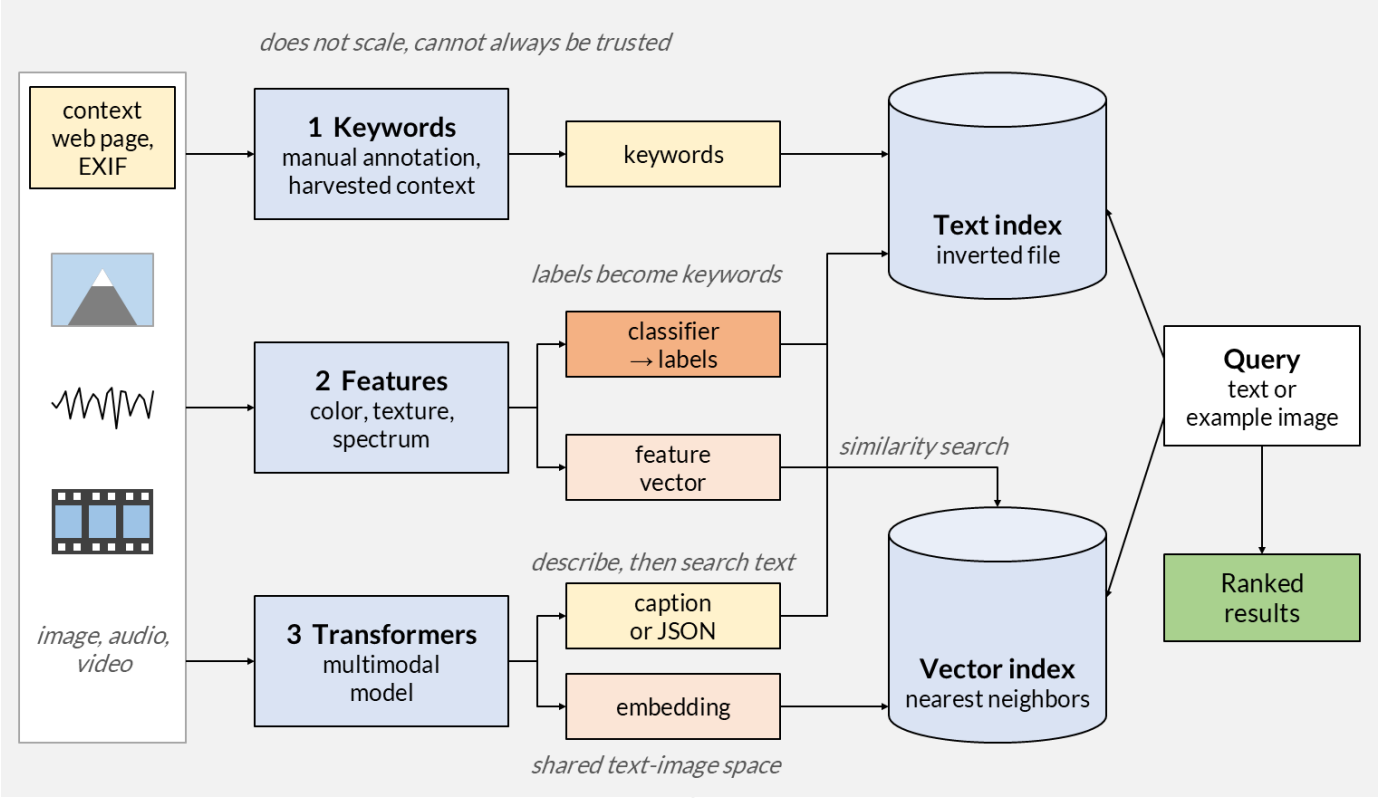
The semantic gap: the photo contains no words

- Chapters 1 to 8: a complete **text toolkit**; now the documents are **pixels, samples, frames**
  - a 1024 × 768 photo: 2,359,296 numbers, none of them "dome"
  - a 3-minute CD song: 15.9 million samples; 1 h of video: 90,000 frames
- **Semantic gap**: computed **features** vs the **concepts** in queries (Smeulders et al., 2000)
- **Two parts**
  - **visible**: "building", "sky", "dome"; closed by content analysis
  - **not in the pixels**: "UNESCO", "Mumtaz Mahal"; needs context (metadata, web page, world knowledge)
- **Not a vocabulary mismatch**: media contain no symbols; their meaning must first be **extracted**



Three paths across the gap

- **1 Keywords**: metadata by humans or harvested (web page, EXIF); reuse the text toolkit
- **2 Features**: color, texture, spectrum
  - **similarity search** on feature vectors
  - **classifiers**: labels feed the keyword path
- **3 Transformers**: multimodal models
  - **describe**: caption or JSON for the text index
  - **embed**: one shared space, vector index
- Plus **time** (audio, video) and **specialized solutions** (song identification)



The keyword path: levels and facets

- **Metadata**: text or structured values describing an item; multimedia search becomes **text search**
  - from humans, capture and harvesting, ML models
- **Facets**: one aspect each; filters and facet counts (chapter 4)
  - "daytime" is visible, "UNESCO" needs context

- **Three levels**
  - **generic**: "building", "sky", "summer"
  - **specific**: "Taj Mahal", "Agra", "2006"
  - **abstract**: "love", "devotion", "iconic"
- Each facet holds values of all levels side by side



Object Facet	Value
Generic Object Instance	building, water, sky
Generic Object Class	mausoleum, tomb, dome, minaret
Specific Named Object Class	UNESCO World Heritage Site (since 1983)
Specific Named Object Instance	Taj Mahal

Spatial Facet	Value
Generic Location	outside
Specific Location Hierarchy	India, Uttar Pradesh, Agra

Event / Activity Facet	Value
Generic Event/Activity	tourism, attraction
Specific Event Instance	International World Heritage Expert Meeting on Visual Integrity in 2006

Temporal Facet	Value
Generic Time	summer, daytime
Specific Time	2006 (photo taken)

Contextual Facet	Value
Topic	Indian Architecture
Related Concepts / Objects	Shah Jehan, Mumtaz Mahal, Islam
Abstract Concept	love, death, devotion, remembrance
Context	built in memory of his favorite wife Mumtaz Mahal, by Shah Jehan; completed 1648

Metadata standards: where, not what

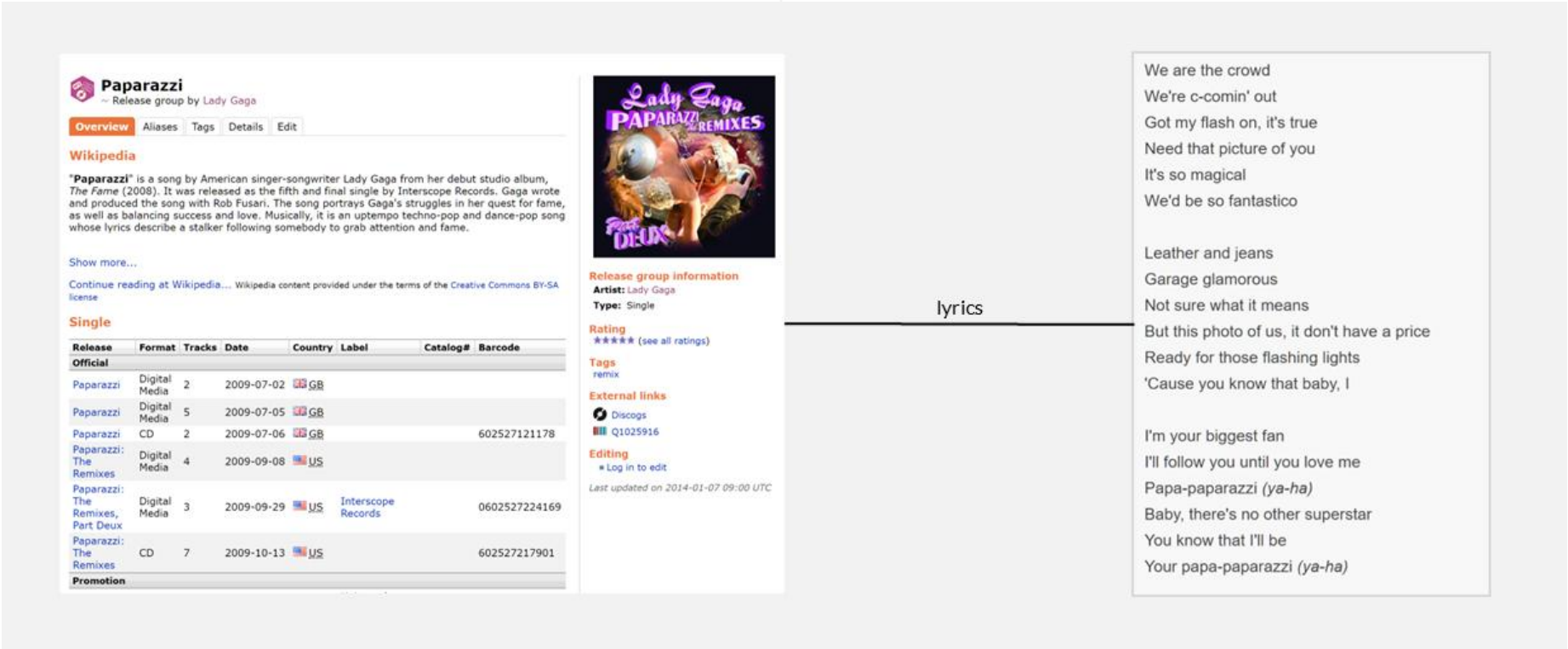
- **EXIF**: written by every camera and phone, unasked
  - timestamp and GPS answer "photos taken in Paris in 2023" without content analysis
  - many platforms **strip it on upload**, mainly for privacy
- **IPTC** (stored as XMP): editorial metadata of the news industry
- **Dublin Core** for libraries, **ID3** in MP3 files, **schema.org** on web pages
  - a VideoObject can list the key moments of a video with timestamps
- Standards fix the **field names**, not the **values**
  - two IPTC keyword fields: "Taj Mahal" vs "religious building"

standard	media	typical fields	written by
EXIF	photos (JPEG, TIFF, HEIF)	date and time, camera, exposure, orientation, GPS	the camera, automatically
IPTC (as XMP)	photos	headline, caption, keywords, creator, location, persons	photographers, agencies
Dublin Core	any resource	15 elements: title, creator, subject, date, rights, ...	libraries, archives
ID3	audio (MP3)	title, artist, album, year, genre, lyrics	labels, software, users
schema.org	web pages (JSON-LD)	ImageObject, VideoObject: name, thumbnail, duration	website owners

Curated catalogues: metadata alone bridges the gap

- **IMDb**: millions of movies, fixed schema
  - cast, crew, dates, genres, plot, ratings
  - maintained by volunteers, crews, studios
- **MusicBrainz**: open music encyclopedia of artists, releases, recordings

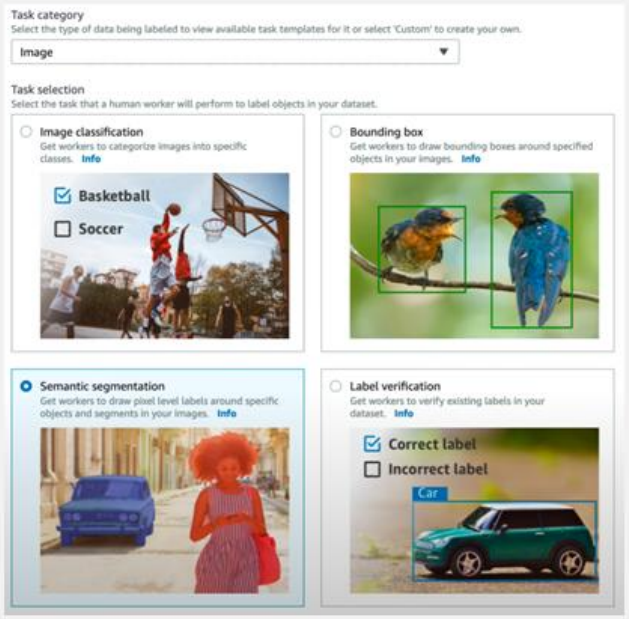
- Search by director or lyrics **never touches the signal**
  - content needed only for "which song is playing?"
- Exists where someone pays: streaming revenue, fan enthusiasm
  - an arbitrary photo collection: **no one pays**



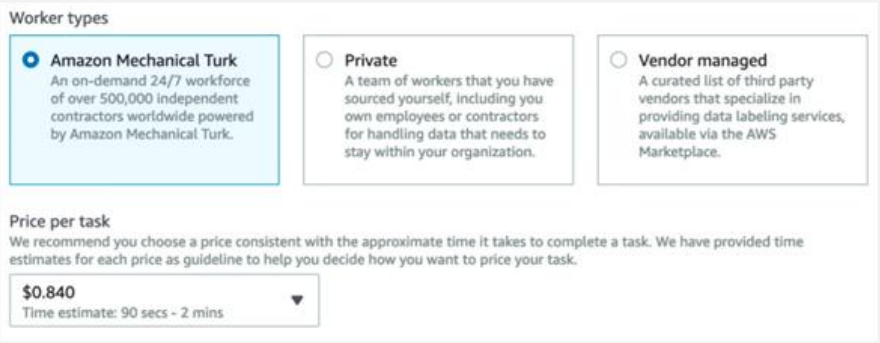
Manual annotation does not scale

- **Crowdsourcing**: Mechanical Turk, over 500,000 workers, fixed price per task
  - Ground Truth: semantic segmentation, 90 s to 2 min per image: **$0.84**
- **Labels are training data too**: the same workforce rated LLM answers
  - a trained model then labels automatically (section 3)

- **1 million photos**, one pass: **$840,000** and **33,333 hours**
  - domain experts cost far more
  - and the archive grows every day
- Only for tasks untrained workers can do, and collections worth the cost





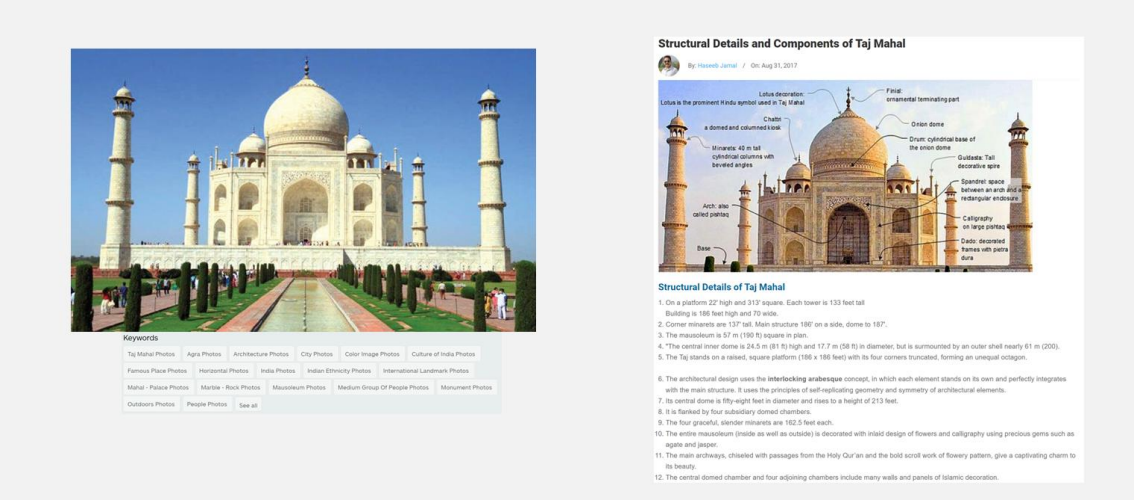


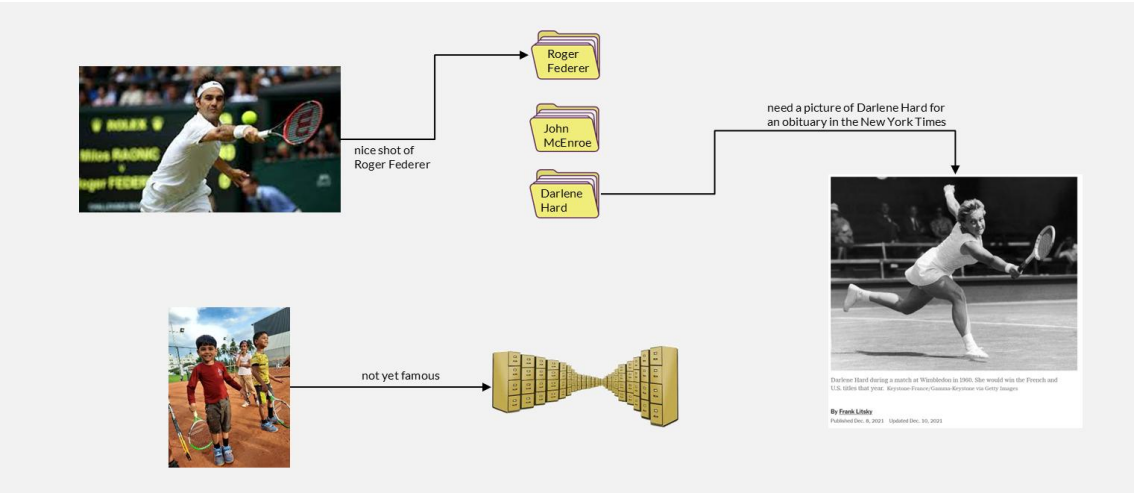
Annotations differ in level and depth

- **Two workers, one photo**: both correct
  - A: "Taj Mahal, India, iconic building, 17th century": specific
  - B: "religious building, great weather, few people outside, nice": generic, abstract
  - embeddings relate the two only to a degree

- **Depth**: stock keywords too thin; expert terms never queried

- **Faceted navigation**: persons, locations, brands, decades
  - Darlene Hard: a curated person folder
  - children playing tennis: unannotated, so unfindable once famous



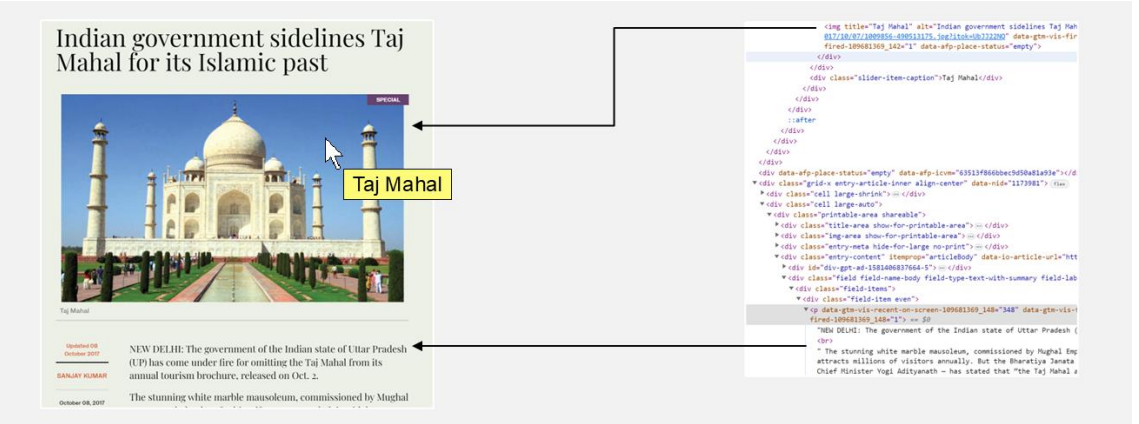


Harvesting context from the web

- Embedded media come with **text at almost no cost**: alt, title, caption, headline, article, page title

- **Which text belongs to the image?**
  - early pages: a token window around the tag
  - modern pages: proximity on the **rendered page** (visual zones)
  - anchor texts of links to the image (chapter 8)

- **Second role: training data**
  - LAION-5B: 5.85 billion image and alt-text pairs
  - cheap, noisy metadata trains CLIP (section 4)





Can we trust metadata? Doctorow's seven obstacles

- Harvested metadata: written by **people with their own goals** (Doctorow, "Metacrap", 2001)
- **A claim, not a fact**: weigh sources by trust, evaluate before use
- Fails on **cost** and **trust**: describe the content from the content

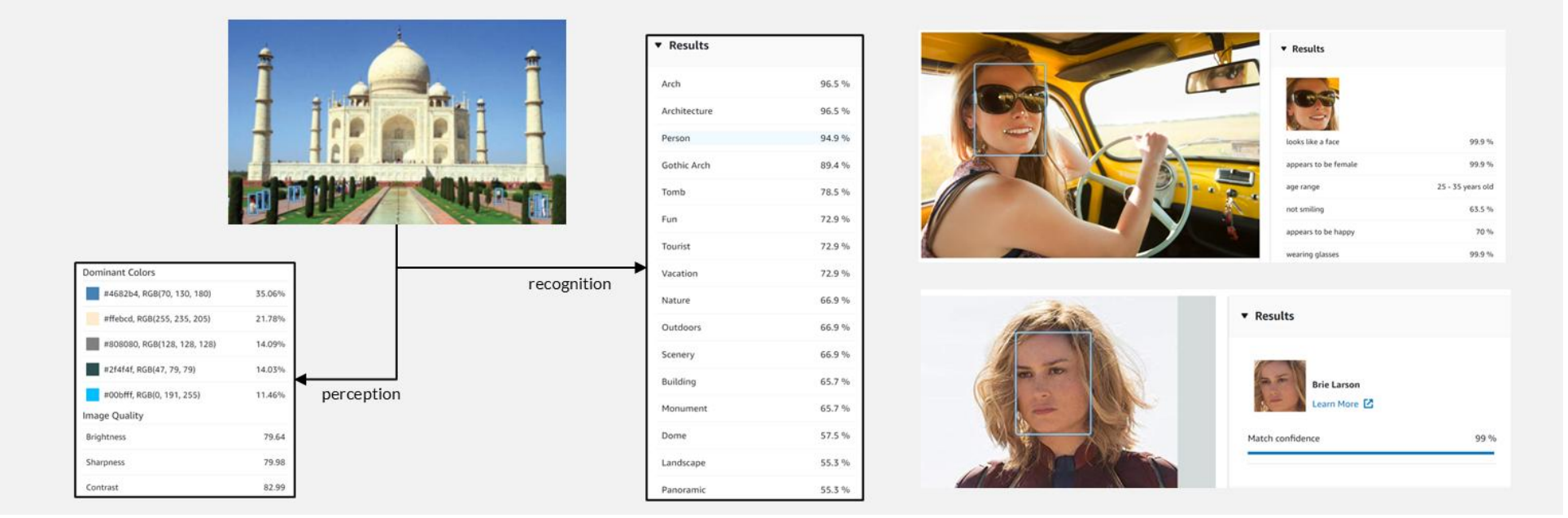
- **Not worthless**
  - curated catalogues: owners care about quality
  - **observational** metadata (anchor texts) is harder to fake, yet Google bombs work

obstacle	in short
1. People lie	misleading metadata attracts traffic
2. People are lazy	many publishers never annotate
3. People are stupid	many lack the skill to catalogue well
4. Mission impossible: know thyself	even honest creators describe their own content inaccurately
5. Schemas aren't neutral	every scheme reflects its designers' views
6. Metrics influence results	what is recorded decides what can be found
7. There's more than one way to describe something	description is subjective; many views are valid

▶ **Demo**: ch09-01-semantic-gap.ipynb

What an image analysis service returns

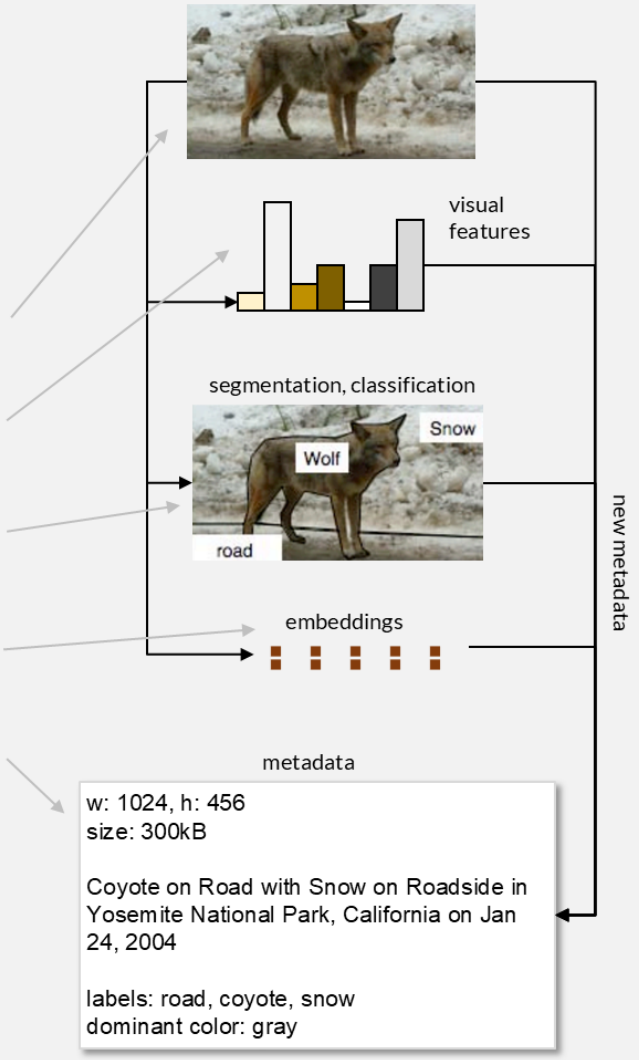
- Taj Mahal photo, nobody annotated it; within a second, two branches
  - **perception**, how it looks: dominant colors (steel blue 35%, light beige 22%, gray 14%), brightness, sharpness, contrast
  - **recognition**, what it shows: Arch 96.5%, Person 94.9%, Tomb 78.5%, Fun 72.9%, Dome 57.5%; face attributes; celebrity (Brie Larson 99%)
- "Gothic Arch" 89.4% for a Mughal building: a **confidence is not a probability of being right**; set a threshold per label from an evaluation (chapter 2)



The abstraction ladder

- Each layer turns the one below into a **more abstract** representation and closes part of the gap

Layer	Examples	Purpose
Raw-Level (Informational)	JPG, MP3, WAV, MP4, TXT	Store original content; basic digital representation
Low-Level (Perceptual)	Color histograms, edge maps, MFCCs, spectral centroid, textures	Capture sensory properties such as color, sound, or motion
Mid-Level (Structural)	Clustering, SVM, CNN/ViT embeddings, object or genre classification	Represent patterns, structure, and relationships within data
High-Level (Semantic)	Captioning, keyword generation, OCR, speech-to-text, topic extraction	Produce interpretable, concept-level meaning
Contextual (Metadata)	Timestamps, geolocation, device data, user tags, social context	Provide external descriptive context for indexing and filtering



Perceptual features: a term vector for images

- **Feature vector** per item: e.g. a color histogram, pixels per color bin

- **Two uses**
  - **similarity search**: query by example, sketch, palette, hummed melody
  - **classification**: labels ("outdoor", "speech") return to the keyword path

- **Invariances**: translation, rotation, scale, lighting and loudness, noise

- Next chapters: the features per modality

text retrieval (chapter 1)	color histogram
term	color bin
term frequency	number of pixels in the bin
document as a term vector	image as a histogram vector
a term refers to a concept	a bin refers to a sensation

Perceptual similarity is not semantic similarity: a red car matches a sunset, while a black cat on a white carpet and a ginger cat on a green lawn do not match

▶ **Demo**: ch09-02-feature-extraction.ipynb

Classification: from features to labels

- **Classic**: hand-crafted features plus a decision tree or SVM

- **Speech vs music** in a radio stream
  - speech pauses: **share of low-energy frames**
  - speech crosses zero irregularly: **ZCR variation**

- Labels become **metadata**: speech to transcription, music to genre; evaluate as in chapter 2

- **Labels are claims**: "Wolf" for the coyote

- **Weak point**: no hand-crafted feature tells "Taj Mahal" from other white domed buildings

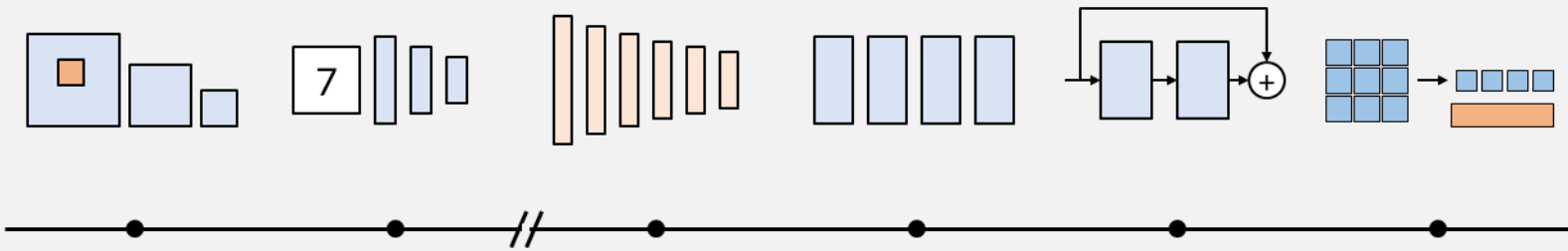
recording, 2 s segments	class	low-energy share	ZCR variation
Munsell (LibriVox)	speech	0.42	0.173
Tucker (LibriVox)	speech	0.37	0.134
Hedrick (LibriVox), test	speech	0.39	0.138
Grieg, Morning	music	0.03	0.025
Grieg, Mountain King	music	0.20	0.015
Bach, Goldberg Aria, test	music	0.14	0.013

```
learned tree (depth 2, trained on 4 recordings):
|--- zcr_variation <= 0.05
|   |--- class: music
|--- zcr_variation >  0.05
|   |--- class: speech

held out: 15 of 15 speech, 15 of 15 music segments correct
```

Neural networks learn the features

- **Learned features** replace hand-crafted ones
  - enablers: ReLU, ImageNet, GPUs, dropout, Adam
  - AlexNet 2012: top-5 error 15.3% vs 26.2%
- **Layers form a ladder**: edges, textures, parts, objects; the second-to-last layer gives the first **learned image embeddings**
- **Today**: self-supervised and contrastive learning, no fixed label sets



1998 Rowley face detector
small net over 20×20 windows, heavy manual preprocessing

1998 LeNet
learned convolutions and pooling, 28×28 digits

2012 AlexNet
learned features beat hand-crafted ones on ImageNet

2014 VGG, GoogLeNet
deeper networks from small reusable blocks

2015 ResNet
residual connections make 100+ layers trainable

2020 Vision Transformer
image as a sequence of 16×16 patches

CLIP and multimodal transformers

today: one pretrained model, no task-specific training

fixed label sets, one training run per task: mostly replaced today

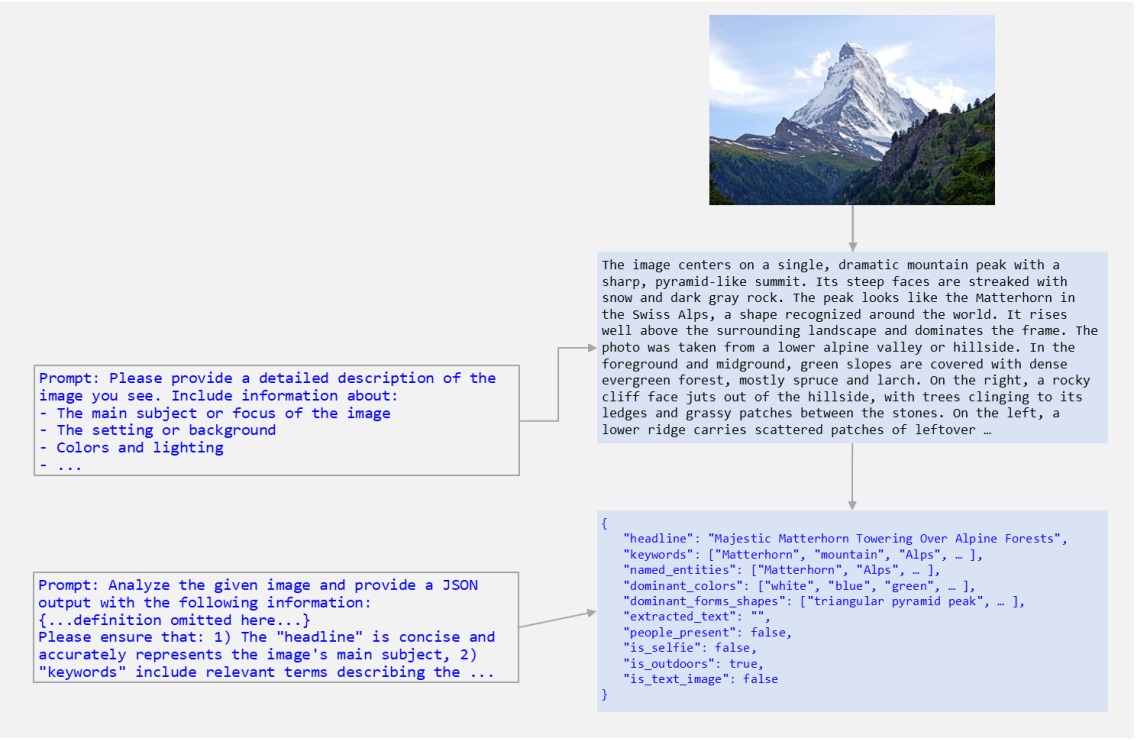
Describe path: generate text, then search it

- **Matterhorn photo**: the model names the peak, without task-specific training

- **Multimodal LLM**: image encoder (ViT) plus language model; the **prompt** controls what it writes
  - two steps: description, then JSON with fixed fields

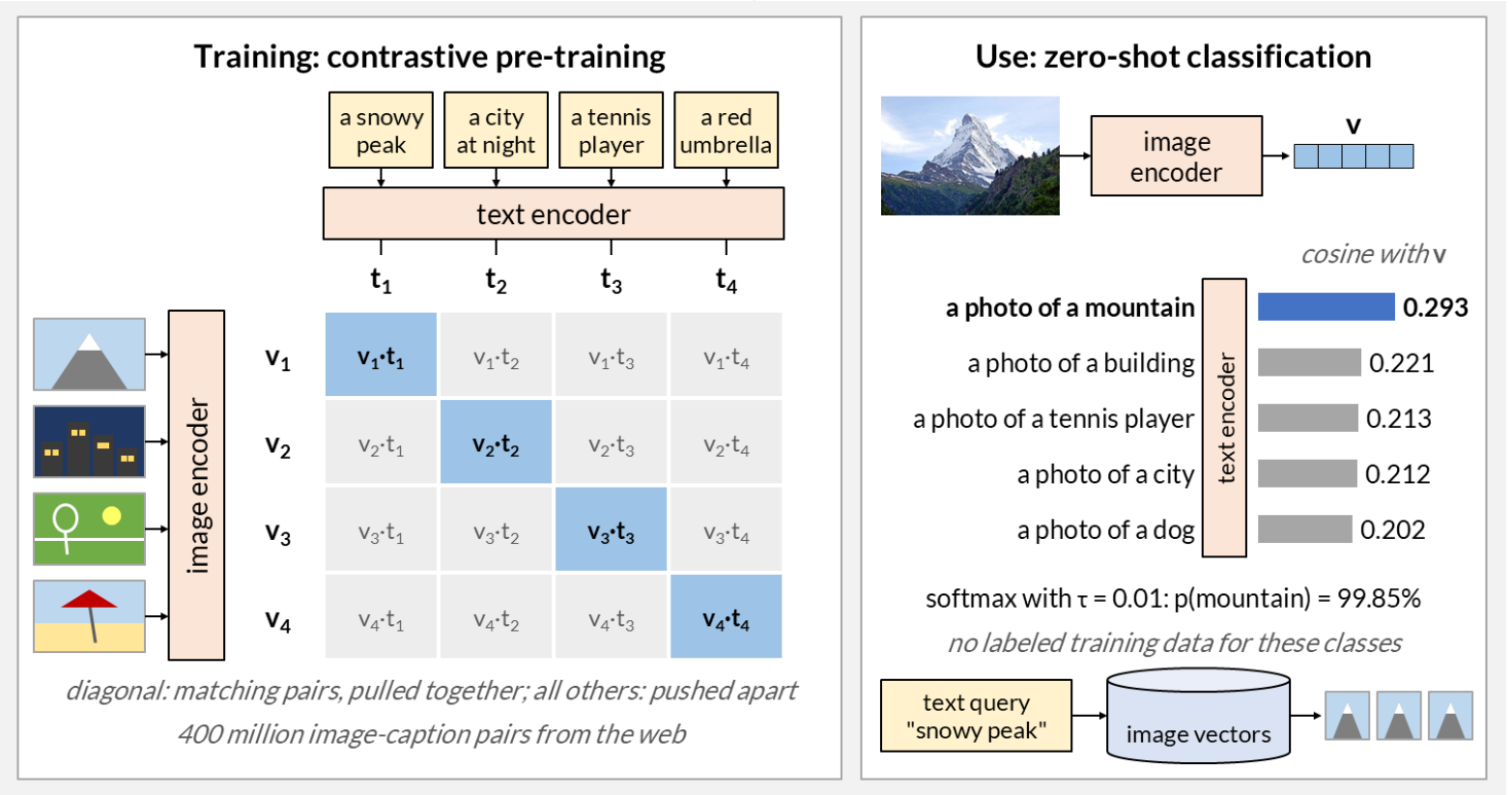
- **Fields into the index** (chapter 4)
  - headline, description: BM25, embeddings
  - keywords, entities: exact match, facets
  - booleans: filters
- New schema = **new prompt**, not retraining

- **Other media**: Whisper transcribes speech with timestamps, OCR reads text in images



Embed path: CLIP puts images and text in one space

- **Two encoders**, one dimension: text transformer, image ViT or CNN
- **Contrastive training** on web image-caption pairs
  - batch of N pairs: N × N cosine matrix
  - diagonal pulled together, rest pushed apart
- No class labels: the **captions** people wrote supervise
- **Retrieval in every direction**: text-to-image via a vector index (chapter 6), image-to-image, image-to-text
- Same idea for audio (CLAP: "dog barking in the rain") and video



Zero-shot classification: the Matterhorn

- **Zero-shot**: one prompt per class ("a photo of a mountain"), pick the closest
- **Softmax over cosine / τ**
  - τ = 0.01, learned: scores = cosine × 100
  - normalize: p = exp / sum of all exp
  - subtract s_max: same p, but no overflow
- **Matterhorn**: mountain 1 / 1.0015 = 99.85%
- **Federer**: tennis player 99.9998%
- **Low cosines**, typically 0.1 to 0.4: the differences carry the information
  - a threshold "cosine > 0.5" rejects every image

$$p(c \mid image) = \frac{e^{s_c}}{\sum_{c'} e^{s_{c'}}} = \frac{e^{s_c - s_{\max}}}{\sum_{c'} e^{s_{c'} - s_{\max}}} \quad with \quad s_c = \frac{\cos(\mathbf{v}, \mathbf{t}_c)}{\tau}$$

prompt	cos	cos / τ	minus max	exp	p = exp / sum
mountain	0.293	29.3	0	1	99.85%
building	0.221	22.1	−7.2	0.00075	0.07%
tennis player	0.213	21.3	−8.0	0.00034	0.03%
city	0.212	21.2	−8.1	0.00030	0.03%
dog	0.202	20.2	−9.1	0.00011	0.01%
sum				1.0015	100%

▶ **Demo**: ch09-03-multimodal-transformers.ipynb

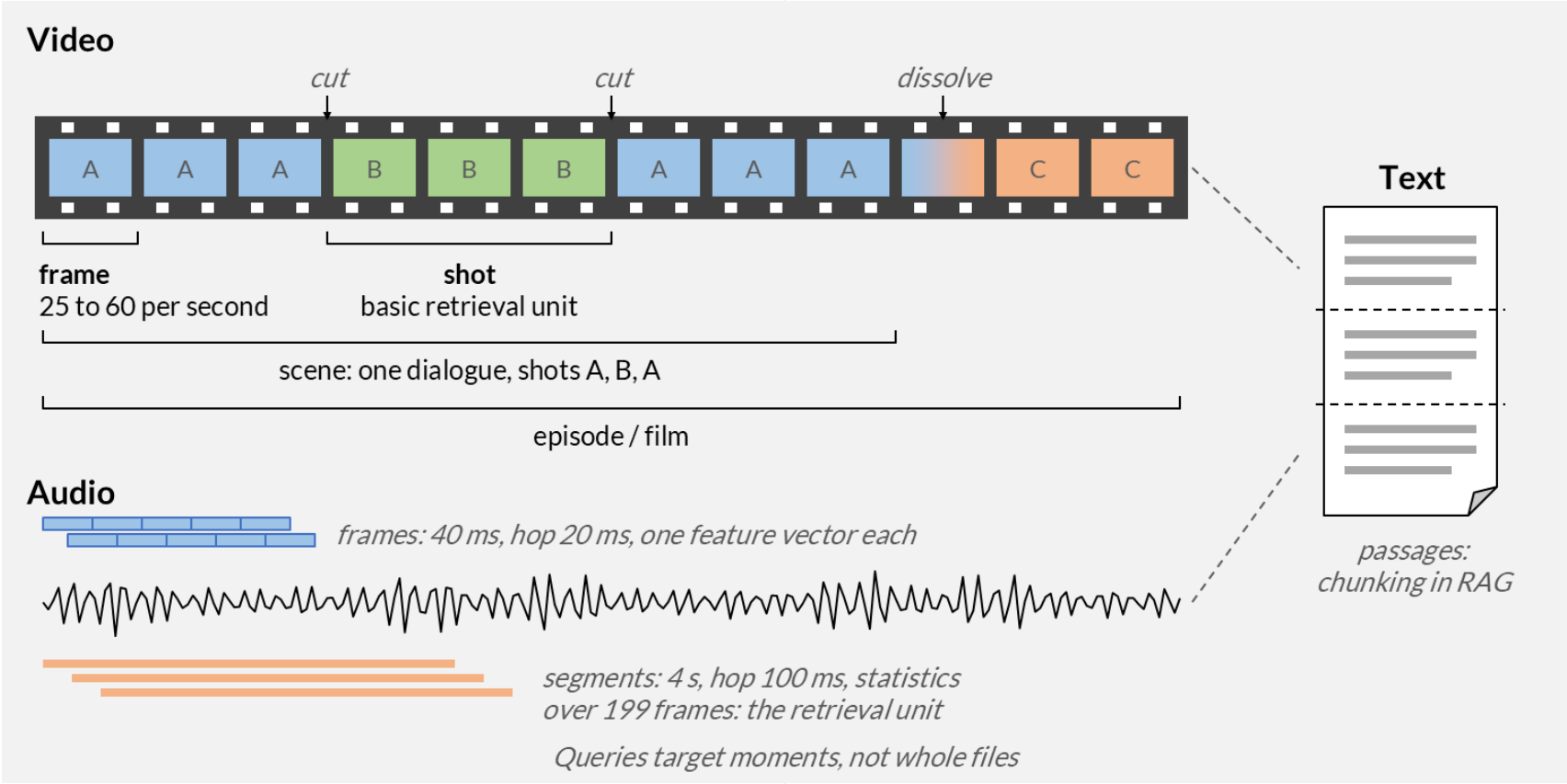
Two transformer paths: describe or embed

- **Two transformer paths**
  - **describe**: captions or JSON in the text index
  - **embed**: images and text in one vector space (CLIP)
- **Describe** for facts and filters: "Matterhorn in winter, no people"
- **Embed** for visual concepts nobody described: "a lonely bench in the fog"
- **Hallucination**, the eighth obstacle: plausible guesses stated with full confidence; worst for names, places, dates
  - cross-check (EXIF GPS), controlled entity lists, mark uncertain values
  - evaluate like a classifier: precision and recall per field
- **Hybrid**: run both paths, fuse with RRF (chapter 5)

aspect	describe path	embed path (CLIP)
index	text index (inverted file), fields, filters	vector index of media embeddings
queries	keywords, names, filters, boolean conditions	free-text descriptions, example images
coverage	only what the description mentions	any visual concept the model learned
explainability	high: matched words are visible	low: a similarity score only
specific entities	good, if the model names them correctly	weak for rare names and exact identifiers
indexing cost	high: a large generative model per item	moderate: one encoder pass per item
typical failure	hallucinated or missing details	plausible but wrong neighbors

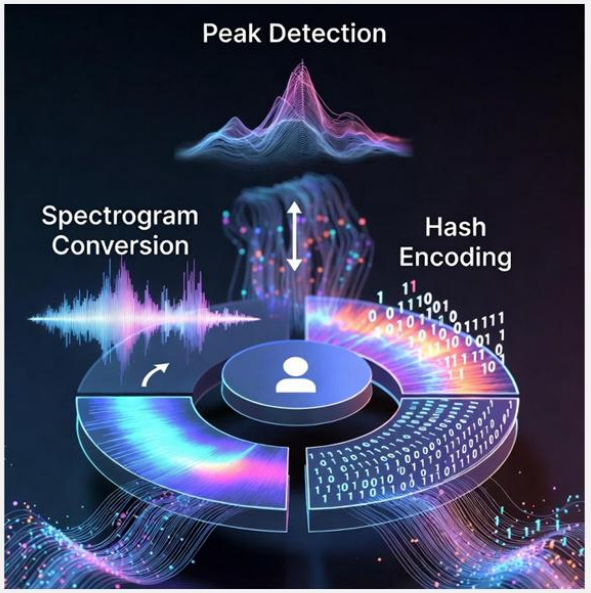
Time-based media need retrieval units

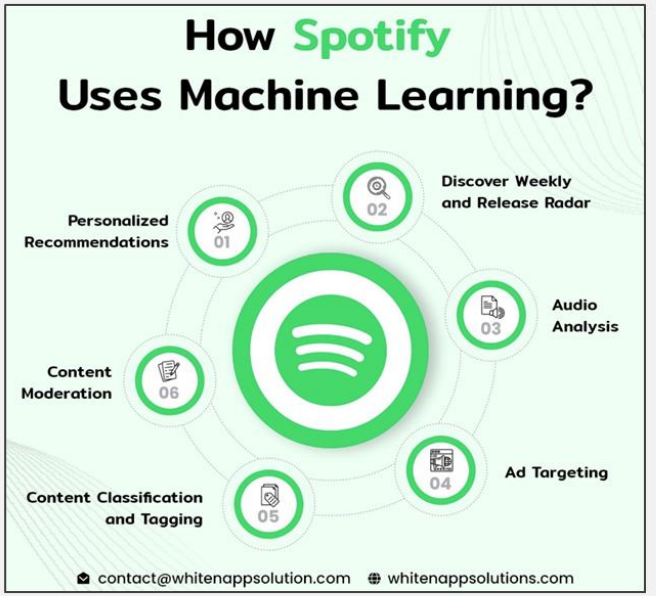
- **Queries** target **moments**, need chunking like in RAG
- **Audio**:
  - overlapping frames (40 ms, hop 20 ms) give features
  - segments (4 s, hop 100 ms) give statistics
- **Video:** 25 to 60 frames per second
  - **shot**: one camera, between cuts; the basic unit
  - **scene**: shots sharing place, time, action
  - **episode**: a film or one part of a series



Specialized solutions: audio fingerprinting

- **Shazam**: 10 s from a noisy bar, millions of recordings, **exactly one** answer within a second
  - noise changes every sample; similarity embeddings return the same genre
- **Fingerprinting**: a robust fingerprint, looked up in a hash index
- **Price**: matches recordings, not songs; singing fails
  - query by humming must compare melodies
- **Opposite**: Spotify combines acoustics, text, and behaviour
- Worth it for "**which exact item** is this?", heavy distortion, huge catalogues
  - also faces, perceptual hashing, copyright detection





Key takeaways

- The semantic gap separates computed features from the concepts in queries; text carries meaning in its symbols, media do not, and part of the gap ("UNESCO World Heritage Site") needs context that the content does not contain
- Metadata turns multimedia search into text search: generic, specific, and abstract values in facets; standards define where metadata lives, not what it says; curated catalogues show the keyword path at its best
- The keyword path breaks for arbitrary collections: annotation costs about a dollar per item and varies between annotators, and harvested metadata can be wrong or manipulated
- Content analysis climbs a ladder from raw data to perceptual, structural, semantic, and contextual levels; perceptual features serve similarity search and classifiers, and perceptual similarity is not semantic similarity
- Neural networks replaced hand-crafted features with learned ones; their labels are claims too and must be evaluated like human metadata
- Multimodal transformers offer two paths: describe the content and reuse the text toolkit (readable, but hallucinations), or embed media and text in one space (CLIP) for text-to-image search and zero-shot labels; hybrid systems use both
- Audio and video need retrieval units such as shots and segments, as text needs chunks; exact tasks such as naming a song from a noisy recording need specialized solutions like fingerprinting

Further reading

- Smeulders et al. (2000). Content-based image retrieval at the end of the early years. IEEE TPAMI.
- Doctorow (2001). Metacrap: Putting the torch to seven straw-men of the meta-utopia.
- Schuhmann et al. (2022). LAION-5B. NeurIPS Datasets and Benchmarks.
- Krizhevsky, Sutskever, Hinton (2012). ImageNet classification with deep CNNs. NeurIPS.
- He et al. (2016). Deep residual learning for image recognition. CVPR.
- Dosovitskiy et al. (2021). An image is worth 16x16 words. ICLR.
- Radford et al. (2021). Learning transferable visual models from natural language supervision (CLIP). ICML.
- Radford et al. (2023). Robust speech recognition via large-scale weak supervision (Whisper). ICML.
- Wang (2003). An industrial-strength audio search algorithm (Shazam). ISMIR.
- IPTC. IPTC Photo Metadata Standard.
- DCMI. DCMI Metadata Terms.
- Schema.org. VideoObject.